

Testing the pecking order theory of capital structure: Evidence from Turkey using panel quantile regression approach

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Abstract

This study tests the validity of the pecking order theory at different investment levels for manufacturing firms listed on the Borsa İstanbul. The study covers the period from 2000 to 2018, and the quantile regression method was employed to determine the relative importance of internal and external funding sources in the financing of firm investments. The empirical findings of the present study reveal that the pecking order theory is valid for the choice behaviour of firms listed on the Borsa İstanbul and that the sensitivity to internal funds and debts increases as investment levels increase. In addition, it is found that small firms act in accordance with the pecking order. The empirical findings show that the pecking order theory is not valid for high- and low-leverage firms; high-leverage firms prefer equity financing at high investment levels when internal funds are insufficient to finance investment expenditures, and low-leverage firms prefer to borrow as their first choice. Moreover, it is found that the firms acted in accordance with the financial hierarchy in the period before the global crisis (2000–2009); in the post-crisis period (2010–2018), debt/borrowing was preferred for investment financing. Finally, firms operating in the food, drink and tobacco product manufacturing; chemical, petroleum, rubber and plastic product manufacturing; and stone- and soil-based industrial product manufacturing subsectors are found to display pecking order behaviour.

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1. Introduction

The pecking order theory is one of the capital structure theories that have been tested in many different economies over the past thirty years. This theory predicts a hierarchy in funding and states in which firms will prefer an internal source of funds to external ones should there be a need for funding. This is the first stage of the pecking order theory. In instances where the needs for funds are not met by internal sources of funding, firms have to choose among external sources of

funds. In this second stage, the theory predicts that firms will prefer low-risk debt from external financing to equity issuance. An examination of the extant literature reveals that the pecking order effect is primarily tested in two ways: the determinants of the target leverage level and the financial deficit regression. Target leverage level models are based on trade-off theory and seek to explain pecking order behaviour through profitability, firm size and growth opportunities (Fama & French, 2002; Rajan & Zingales, 1995; Titman & Wessels, 1988). Models based on the financial deficit regression are developed as alternatives to the trade-off model and seek to explain the pecking order effect by including all components of a financial deficit (i.e., capital expenditures and dividend payments) as external variables to the model provided that debt can be securely attained (Frank & Goyal, 2003; Leary & Roberts,

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2010; Shyam-Sunder & Myers, 1999). Irrespective of the extent to which these models provide information about the financial behaviour of firms, they do not provide information about firms' capital expenditures or their financial preferences that change over time. Chay, Park, Kim, and Suh (2015) introduced the investment regression model by developing the cash flow sensitivity model of Fazzari, Hubbard, and Petersen (1988) to investigate the financial behaviour of firms that changes with the investment level. With a model based on the quantile regression, pecking order behaviour taking into account the changes in the capital expenditure level of firms was examined using a sample of developed countries. The results of the research showed that firms operating in the US and six developed stock exchanges have higher sensitivity to domestic investment in the first rung of the pecking order, and the sensitivity to stock issuance is higher in the second rung of the pecking order compared to debt financing. Although this model gives good results, the quantile regression method being utilized ignores endogeneity problems and firm-specific differences. In addition, although the study covers a large data set, it does not provide information about the financial behaviour of firms in developing countries. Emerging markets differ from developed markets in their investor characteristics, institutional quality, the development of debt and equity issuance markets. The purpose of this study is to investigate the presence of a financial hierarchy hypothesis at different levels of investment. Firms can make investment decisions at different levels from time to time. In general, they finance their low-cost investments with internal funds; however, when financing medium- or high-cost investments, firms resort to external funds since internal funds are mostly insufficient to meet the required level of funds. The focus of the study is to investigate to what level firms use internal funds, after which investment level they resort to external funds and whether they meet their external funding needs primarily through borrowing or equity issuance.

The study is conducted using manufacturing firms listed on the Borsa İstanbul (formerly named the İstanbul Stock Exchange until 2013). Investigating the pecking order effect on firms in Turkey within an emerging market context will provide a suitable environment to test this hypothesis. Just as it is experienced or seen in other developing countries, Turkey also has a banking sector-oriented borrowing market. Additionally, the intense use of domestic public debt instruments prevents the development of private sector borrowing markets and limits the low-cost borrowing alternatives available to firms through bond issuance. Hence, like other emerging markets, foreign investors encounter serious problems in this market compared to the markets of developed countries; and firms are expected to exhibit pecking order behaviour due to lack of information, information management, problems associated with profit sharing, limited legal rights, and long and costly processes to seek justice in the judicial system. This study solves the endogeneity problem in the investment quantile regression model proposed by Chay et al. (2015) using the fixed effect in the location-scale approach developed by Machado and Silva (2019). Besides, in the quantile regression

model used by Chay et al. (2015), individual effects were not considered. The model used in this study allows for the use of methods that are valid for the prediction of conditional tools, such as differentiating individual effects in panel data models, while also providing information on how regressors affect the entire conditional distribution.

This study examined pecking order behaviour in different directions using 2197 observations from 179 manufacturing firms listed on the Borsa İstanbul from 2000 to 2018. First, this study investigated the existence of pecking orders at different investment levels using all the firms in the sample. In the second stage, following the differentiation of pecking order behaviour in large and small firms in other studies (Booth, Aivazian, Demircuc-Kunt, & Maksimovic, 2001; Huang, 2006; Rajan & Zingales, 1995), firms are classified according to their sizes and pecking orders into large and small firms, and their pecking order differentiation is examined. In the third stage, this paper considered how changes in economic conditions could shape the financial behaviour of firms and investigated how the pecking order effect changed in the periods from 2000 to 2009 and 2010 to 2018. In the fourth stage, since the risk level of the firms has an important effect on financing new investments, the sample is divided into two groups as high risk/leveraged and low risk/leveraged firms, and the effect of the pecking order is examined. Finally, the pecking order behaviour of firms is determined within manufacturing industry subsectors since the subsectors may impact the financial behaviour of the firms since they may have different characteristics.

The model's results provide evidence that firms within the manufacturing industry listed on the Borsa İstanbul act in accordance with the pecking order theory. It was observed that firms preferred internal funds to external funds in the first rung of the pecking order; in the second rung, they turned to debt financing. Moreover, as the investment level increases, firms' sensitivity to internal funds and debts also increases. It was found that large firms have higher sensitivity to internal fund investments compared to small firms, but small firms act in accordance with the financial hierarchy. However, when the pecking order effect was not continuous, there was evidence that firms turned to external financing from 2010 to 2018. In addition, it was found that high-leveraged firms prefer internal funds to external funds, and, second, they prefer to issue shares when investment levels are high while low-leverage firms primary prefer to issue debt. According to the results of the analysis of the subsectors, it was observed that firms show pecking order behaviour in the food, drink and tobacco product manufacturing; chemical, petroleum, rubber and plastic product manufacturing; stone- and ground-based industrial product manufacturing subsectors. The rest of the paper is organized as follows. In section two, the related literature about the pecking order theory is reviewed. Detailed information about the data set and the methodological background are presented in section three. Section four introduces the empirical analysis and presents the findings, and the paper concludes with discussions and recommendations related to the empirical findings.

2. Related literature

Discussions on capital structure in the financial literature continue unabated. While many theories advocate that there is an optimal capital structure for firms, other studies have provided evidence that this view cannot be supported. [Modigliani and Miller \(1958\)](#) claimed that capital structure would not affect firm value under certain conditions. In later periods, two main views dominated discussions on capital structure in the existing literature. The first of these views is the trade-off theory, which integrates financial distress and agency costs with earlier models proposed by [Modigliani and Miller \(1963\)](#) and [Miller \(1977\)](#). Trade-off theory aims to balance the bankruptcy and distress costs associated with the use of leverage with the tax benefits from using debt and a reduction in the agency costs that this creates. The other supported view on capital structure is the pecking order theory proposed by [Myers \(1984\)](#) and [Myers and Majluf \(1984\)](#), which is in line with the financial hierarchy for firms' long-term financing strategies initially suggested by [Donaldson \(1961\)](#). This theory advocates that firms use internal sources, debts and share issuances in a certain order to meet their funding needs. The reason firms exhibit such behaviour stems from the information asymmetry between managers and shareholders. [Shyam-Sunder and Myers \(1999\)](#) developed an effective model based on financial deficits to test pecking order behaviour. With this model, they studied 157 firms in the US in the period from 1981 to 1989. They explained changes in financial deficits and changes in the leverage ratio and found strong evidence that large firms follow the pecking order theory. However, [Frank and Goyal \(2003\)](#) stated that although the financial deficit model is an important factor in explaining the leverage of a company, it could not provide evidence supporting the pecking order theory because asymmetric information is more effective in small public firms.

[Fama and French \(2002\)](#) applied a triple regression model, investigated the consistency of trade-off and pecking order estimates in their study and tested the relation between dividend and leverage according to various firm characteristics. The results of the study showed that profitable firms that invest less acted in accordance with the pecking order theory, and they found positive results supporting the theory between firm size and dividend payments and leverage. [Leary and Roberts \(2010\)](#) stated that the model developed by [Shyam-Sunder and Myers \(1999\)](#) could not sufficiently explain the hierarchical structure of the pecking order theory and proposed a different model. The results of the corresponding model showed that 67% of firms prefer internal funds, 23% prefer debt financing and 10% prefer equity financing. They also argued that equity financing is especially important for small and newly established firms. [Chay et al. \(2015\)](#), in an attempt to study the financial behaviour of firms that changes with investment levels, expanded the cash flow sensitivity model developed by [Fazzari et al. \(1988\)](#) and introduced the investment regression model. The results of this model, which is based on a quantile regression, revealed that firms operating in the US and six developed stock exchanges had higher

sensitivity in domestic financing at all investment levels in the first rung of the pecking order while the sensitivity to stock issuance was higher than debt financing in the second rung of the pecking order.

A review of the extant literature reveals that researchers agree on the first rung of the pecking order theory; thus, firms prefer internal finance to external funds ([Chay et al., 2015](#); [Fazzari et al., 1988](#); [Hoshi, Kashyap, & Scharfstein, 1991](#)). However, there is no consensus on the second rung of the pecking order in the existing literature. [De Haan and Hinloopen \(2003\)](#) also included bank loans in their model and found that the pecking order for some selected Dutch firms were internal sources of funds and then bank loans. If there is a need for more funds, firms prefer to issue shares and issue bonds in that order. [Anderson and Carverhill \(2012\)](#) stated that the optimal cash holding policy may lead to a financial hierarchy depending on the business conditions (state-dependent hierarchy). In another study, [Dang \(2013\)](#) tested the pecking order theory of capital structure for firms in the UK, Germany, and France and concluded that the capital structure decisions of the firms did not follow this theory. In particular, the results of the study stated that these firms used debts to offset a small portion of the financing deficit or surplus.

In most of the studies on developing countries, the pecking order effect has been tested with an emphasis on the determinants of the target debt levels based on trade-off theory and financial deficit models. The financial deficit regression model has frequently been tested in several different economies including Taiwan ([Chen, Chen, Chen, & Huang, 2013](#)), India ([Komerla and Lukose, 2015](#)) and Egypt ([Allini, Rakha, McMillan, & Caldarelli, 2018](#)), despite there being no evidence supporting the pecking order effect. [Seifert and Gonenc \(2010\)](#) found a similar result in a study covering 23 developing countries and stated that in these economies, issuing stock was the first referral source for firms needing funds. In addition, when the studies using the target debt determinants model are examined, [Booth, Aivazian, Demircug-Kunt, and Maksimovic \(2001\)](#), using data from developing countries (Brazil, Mexico, India, South Korea, Jordan, Malaysia, Pakistan, Thailand, Turkey and Zimbabwe), found that firms avoid costly external financing and exhibit behaviour in line with the first tier of the pecking order. In a related study on Chinese firms, [Chen \(2004\)](#) found that domestic funds were the first referral source for firms, and equity financing was preferred to debt financing. [Chakraborty \(2010\)](#), in his study on Indian firms, stated that in the post-economic reform period, firms displayed strict pecking order behaviour, especially firms tending to borrow from financial institutions after internal financing options were exhausted. [Oino and Ukaegbu \(2015\)](#) found that Nigerian firms display a certain amount of pecking order behaviour. Almost all of the research on the pecking order theory in the literature on Turkey tested the determinants of target debt levels. Most of the studies yielded results that support the pecking order theory in Turkish firms ([Abdioğlu & Deniz, 2015](#); [Doğukanlı & Acaravci, 2004](#); [Durukan, 1997](#); [Onatca Engin, Unver Erbas, & Sokmen, 2019](#)). Köksal and

Orman (2015) found evidence that pecking order behaviour is observed in small firms operating in the manufacturing industry, especially in unstable environments, although they are not as strong as the trade-offs in their study on Turkish firms.

As seen in the existing literature, there are different opinions regarding the validity of the pecking order theory in developing economies. However, these studies are insufficient in terms of the financial behaviour of firms at different investment levels because besides the target debt ratio, the amount of investment expenditures also affects the financing decisions of firms. In this context, our study investigates whether firms operating in a developing economy follow pecking orders in their financial decisions according to their investment levels. In addition, the study contributes to the literature by investigating the effects of firm risk, firm size, sectoral differences and period differences in financial preferences that vary according to the investment levels of firms. Additionally, using a data set covering a long period and using the location-scale approach quantile regression model, which takes into account individual effects, are among the important differences of the study.

3. Data set and methodology

The study covers 2197 observations of 179 firms operating in the Borsa İstanbul manufacturing industry from 2000 to 2018. Sectoral distributions and the numbers of observations of the firms included in the data set are given in Table 1. While determining the data set in the study, all sectors except the financial sector and holdings were targeted; furthermore, since the scope of the study was limited to the manufacturing industry due to the small number of firms and the separation into different subsectors in the service, transportation, construction and technology sectors, they were not included in the analysis. In addition, in periods where capital expenditure figures were negative because firms did not undertake investment activities or sold tangible assets, such firms were excluded from the sample. Additionally, negative values were observed for the internal and external funds of firms in some years. Since negative values did not represent a fund entry, they were also excluded from the sample. The dataset for this study was extracted from the Finnet database on firms operating within the Turkish manufacturing industry and listed on the Borsa İstanbul.

Table 1
Manufacturing industry sub-sectors of this study.

Manufacturing industry sub-sectors	Firm	Observations
Food, drink and tobacco products manufacturing	29	296
Weaving, clothing and leather products manufacturing	24	273
Forest and forest products manufacturing	18	217
Chemical, petroleum, rubber and plastic products manufacturing	30	356
Stone and soil based industrial products manufacturing	28	445
Metal main industry products manufacturing	19	236
Metal goods and machinery products manufacturing	31	374
Total manufacturing industry	179	2197

Table 2
Descriptive statistics.

Variables	All	1	2	3	4	5	6	7	8	9	10
<i>I/K</i>	0.278 (0.229)	0.031 (0.014)	0.071 (0.010)	0.110 (0.010)	0.145 (0.010)	0.187 (0.013)	0.241 (0.017)	0.305 (0.022)	0.403 (0.029)	0.524 (0.040)	0.770 (0.136)
<i>CF/K</i>	0.366 (0.323)	0.284 (0.303)	0.294 (0.260)	0.323 (0.277)	0.361 (0.311)	0.348 (0.255)	0.365 (0.303)	0.365 (0.316)	0.419 (0.376)	0.435 (0.348)	0.466 (0.410)
<i>M/B</i>	1.834 (2.567)	1.432 (1.526)	1.494 (1.491)	2.236 (5.340)	1.741 (2.523)	1.471 (0.938)	1.923 (2.258)	2.073 (2.600)	2.104 (2.263)	2.164 (2.292)	1.714 (1.667)
<i>EXTF/K</i>	0.216 (0.547)	0.111 (0.262)	0.114 (0.266)	0.137 (0.327)	0.175 (0.481)	0.159 (0.394)	0.231 (0.732)	0.233 (0.430)	0.280 (0.460)	0.369 (0.638)	0.351 (0.982)
<i>DEBT/K</i>	0.189 (0.451)	0.081 (0.191)	0.108 (0.260)	0.125 (0.325)	0.167 (0.477)	0.134 (0.296)	0.222 (0.721)	0.196 (0.305)	0.244 (0.433)	0.331 (0.590)	0.281 (0.557)
<i>EQUITY/K</i>	0.026 (0.244)	0.030 (0.188)	0.005 (0.042)	0.012 (0.059)	0.007 (0.083)	0.025 (0.273)	0.008 (0.055)	0.037 (0.320)	0.030 (0.143)	0.036 (0.275)	0.069 (0.528)
<i>LN(Sales)</i>	19.217 (1.708)	18.650 (1.362)	19.338 (1.423)	19.370 (1.616)	19.552 (1.673)	19.444 (1.711)	19.975 (1.715)	19.674 (1.918)	19.132 (1.844)	18.500 (1.616)	18.533 (1.457)
<i>Cash/K</i>	0.354 (0.603)	0.320 (0.533)	0.373 (0.664)	0.364 (0.603)	0.444 (0.760)	0.273 (0.457)	0.366 (0.643)	0.319 (0.473)	0.421 (0.713)	0.359 (0.552)	0.298 (0.541)
<i>Leverage</i>	0.436 (0.199)	0.366 (0.206)	0.386 (0.186)	0.400 (0.203)	0.430 (0.187)	0.422 (0.185)	0.472 (0.183)	0.495 (0.202)	0.458 (0.197)	0.493 (0.195)	0.435 (0.201)

Notes: Descriptive statistics are given for the whole sample (all) covering the period 2000–2018 and for the decile subgroups based on investments (*I/K*). The table shows the mean and their associated standard deviations in parenthesis.

This study is based on a regression model by Fazzari et al. (1988) used to analyze differences in firms' investment behaviour. This model is formulated as follows:

$$(I/K)_{it} = \beta_{CF/K}(CF/K)_{it} + \beta_{M/B}(M/B)_{it} + u_{it} \quad (1)$$

where i and t denote the firm and time variables, respectively. The dependent variable is (I/K) , where I denotes the capital expenditures in the relevant period, and K denotes net fixed assets. (CF/K) is the explanatory variable, where (CF) represents the current period's cash flow as measured by net income plus depreciation plus the change in deferred taxes scaled by K (Chay et al., 2015; Cleary, 1999). (M/B) is the sole control variable, and it denotes the market-to-book ratio. Finally, the coefficient denotes investment-cash flow sensitivity. Chay et al. (2015) test the first rung of the pecking order by including external funds in the model proposed by Fazzari et al. (1988) in Equation (1). This model is used to seek answers to the questions that firms face while deciding whether to prioritize internal sources of funding or external sources of funding to finance their investment activities. The coefficient values of the established model are used to measure firms' funding preferences. The model is specified as follows:

$$(I/K)_{it} = \beta_{CF/K}(CF/K)_{it} + \beta_{EXTF/K}(EXTF/K)_{it} + \beta_{M/B}(M/B)_{it} + u_{it} \quad (2)$$

The $(EXTF/K)$ variable in Equation (2) is the explanatory variable representing external fund usage. $EXTF$ is formed from the sum of the increment in firms' financial debts (bank credits and bond issuances) with equity issuances. Firms in Turkey do not mostly resort to bond issuances as a source of debt financing due to high rates of inflation and interest. Therefore, bank loans comprise the majority of their financial debts. In Equation (2), the coefficients depict the importance of the internal and external funds in investment finances. Chay et al. (2015) proposed another model to test the second rung of the pecking order theory. In their model, they separated the $(EXTF/K)$ variable, which represents the use of external funds in Equation (2), into two groups: $(DEBTF/K)$ and $(EQUITYF/K)$.

$(DEBTF/K)$ represents the debts of firms, and $(EQUITYF/K)$ denotes equity issuances. The proposed model is formulated as follows:

$$(I/K)_{it} = \beta_{CF/K}(CF/K)_{it} + \beta_{DEBTF/K}(DEBTF/K)_{it} + \beta_{EQUITYF/K}(EQUITYF/K)_{it} + \beta_{M/B}(M/B)_{it} + u_{it} \quad (3)$$

The $(DEBTF)$ variable in Equation (3) shows the change in the financial debts of firms, that is, in their total long-term and short-term financial debts. $(EQUITYF)$ shows the total equity issuances of the firms in the related period. Some control variables can be added to Model 2 and Model 3 and re-estimated. $\ln(\text{sales})$ represents the firm's size, and it is calculated by taking the natural logarithm of the sales. (Cash/K) shows the cash flow of the firms, and it is calculated by dividing the firm's net cash assets for the relevant period by net tangible assets. The leverage variable represents the capital structure of the firms and has been calculated by dividing the total debt by the total assets. There are a number of studies in the extant literature that used panel quantile regression models to measure the pecking order effect. Quantile regression models allow the researcher to account for unobserved heterogeneity and heterogeneous covariate effects, and the availability of panel data potentially allows the researcher to include fixed effects to control for some unobserved covariates (Canay, 2011). Unlike the traditional quantile regression model, which provides for the effects of the regressors at different levels of the dependent variable (Qamruzzaman & Wei, 2019), the panel quantile regression model considers unobserved individual heterogeneity and distributional heterogeneity. One of the most important advantages of using this model is that it shows the common effect of fund preferences that vary across firms at different investment levels.

The location-scale approach developed by Machado and Silva (2019) was used in the study. The approach was used for two reasons. First, this model has been developed as an alternative to dynamic quantile panel data models because it is much simpler to calculate and can be applied to nonlinear or polynomial models as with other estimators such as the

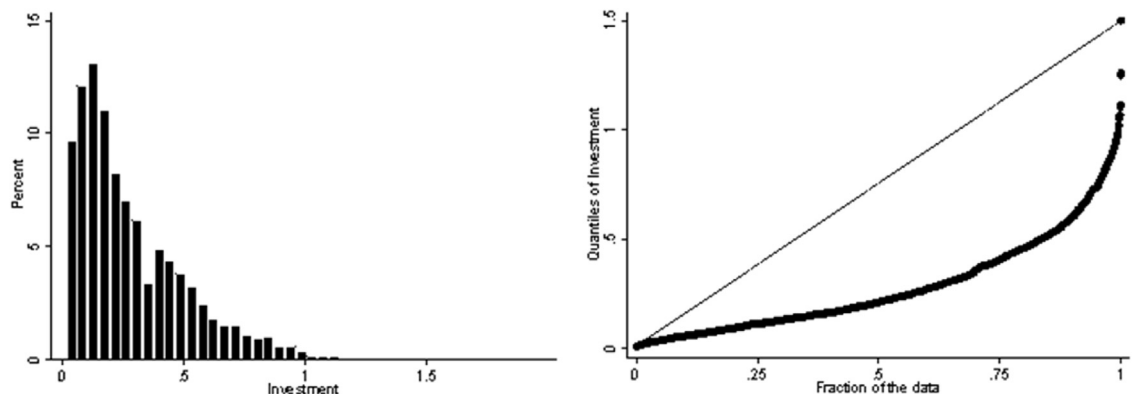


Fig. 1. Distribution of investments.

instrumental variable quantile regression estimator (IVQR). Second, it allows for fixed effects, as in this case. The fixed effects account for some form of endogeneity in the sense that they account for omitted variables that are constant over time and that can be correlated with the regressors. Some advantages of this approach are as follows: First, this approach allows the use of methods that are valid for the prediction of conditional tools, such as differentiating individual effects in panel data models, while also providing information on how regressors affect the entire conditional distribution. Perhaps the most attractive feature of the quantile regression is its ability to provide this information. Second, this approach greatly facilitates the prediction of complex models, as well as estimates of regression values that do not exceed a certain threshold, which is an important requirement that is often overlooked in empirical applications. Finally, this approach is not based on the estimation of the conditional means, but rather it is based on the moment conditions that exogenously identify conditional means. Therefore, the approach can also be adapted to the estimation of cross-sectional models with endogenous variables. This is especially applicable to nonlinear models and is computationally much simpler, especially in models with multiple endogenous variables (Machado & Silva, 2019).

Following Machado and Silva (2019), let $\{(I/K)_{it}, (X'_{it})'\}$ be the data set, where $(I/K)_{it}$ denotes the investments (I/K) in firm i at time t and X_{it} represents internal funds (CF/K) , external funds $(EXTF/K)$, debt financing $(DEBTF/K)$, equity financing $(EQUITYF/K)$ and the control variables in firm i at time t . The estimation of the conditional quantiles of investments $Q_{(I/K)}(\tau|X)$ uses a location-scale model of the following form:

$$(I/K)_{it} = \alpha_i + X'_{it}\beta + (\delta_i + Z'_{it}\gamma)U_{it} \quad (4)$$

with $P\{(\delta_i + Z'_{it}\gamma) > 0\} = 1$. The parameters $(\alpha_i, \delta_i), i = 1, \dots, n$, capture the fixed effects of firm i , and Z is the known differentiable transformations of the components of X . β denotes a vector of estimated parameters in the equation, which vary for different quantiles τ of investments. The sequence $\{X_{it}\}$ is i.i.d. for any fixed firm i and independent across time t . U is an unobserved random variable; and U_{it} is i.i.d. across firm i at time t , statistically independent of X_{it} , and normalized to satisfy the moment conditions:

$$E(U) = 0, E(|U|) = 1 \quad (5)$$

Therefore, the panel quantile functions for the first rung and second rung of the pecking order are specified as follows:

$$Q_{(I/K)_{it}}(\tau|\alpha_i, \varepsilon_t, X_{it}) = \alpha_i + \beta_{1\tau}(CF/K)_{it} + \beta_{2\tau}(EXFT/K)_{it} + \beta_{3\tau}(M/B)_{it} + \varepsilon_{it} \quad (6)$$

$$Q_{(I/K)_{it}}(\tau|\alpha_i, \varepsilon_t, X_{it}) = \alpha_i + \beta_{1\tau}(CF/K)_{it} + \beta_{2\tau}(DEBTF/K)_{it} + \beta_{3\tau}(EQUITYF/K)_{it} + \beta_{4\tau}(M/B)_{it} + \varepsilon_{it} \quad (7)$$

Table 3
Pecking order quantile regression results in the first rung.

Variables	FE	.10	.20	.30	.40	.50	.60	.70	.80	.90
Panel A: with M/B as a control variable										
CF/K	0.090 (0.019)***	0.084 (0.023)***	0.085 (0.020)***	0.086 (0.019)***	0.087 (0.017)***	0.088 (0.017)***	0.090 (0.019)***	0.092 (0.026)***	0.096 (0.037)***	0.100 (0.053)*
EXTF/K	0.054 (0.009)***	0.035 (0.020)*	0.039 (0.017)**	0.041 (0.016)**	0.044 (0.015)***	0.047 (0.014)***	0.053 (0.016)***	0.060 (0.022)***	0.069 (0.032)**	0.081 (0.046)*
M/B	-0.002 (0.002)	0.001 (0.002)	0.001 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	-0.002 (0.001)	-0.004 (0.002)*	-0.006 (0.003)*	-0.009 (0.004)*
Number of obs.	2197									
F-Value	20.51***									
Panel B: with additional control variable										
CF/K	0.121 (0.018)***	0.151 (0.026)***	0.145 (0.022)***	0.140 (0.020)***	0.134 (0.018)***	0.128 (0.018)***	0.120 (0.020)***	0.110 (0.026)***	0.101 (0.034)***	0.081 (0.049)*
EXTF/K	0.035 (0.009)***	0.028 (0.019)	0.030 (0.016)*	0.031 (0.014)**	0.032 (0.013)**	0.033 (0.013)**	0.035 (0.014)**	0.037 (0.018)**	0.039 (0.025)	0.043 (0.035)
M/B	-0.005 (0.002)***	-0.000 (0.002)	-0.001 (0.002)	-0.002 (0.001)	-0.003 (0.001)*	-0.004 (0.001)**	-0.005 (0.001)***	-0.006 (0.002)***	-0.008 (0.003)***	-0.010 (0.004)***
Log(Sales)	-0.108 (0.005)***	-0.086 (0.007)***	-0.091 (0.006)***	-0.095 (0.005)***	-0.099 (0.005)***	-0.103 (0.005)***	-0.108 (0.006)***	-0.115 (0.007)***	-0.123 (0.010)***	-0.136 (0.014)***
Cash/K	-0.046 (0.009)***	-0.037 (0.013)***	-0.039 (0.011)***	-0.040 (0.010)***	-0.042 (0.009)***	-0.044 (0.009)***	-0.047 (0.010)***	-0.050 (0.013)***	-0.053 (0.017)***	-0.059 (0.024)***
Leverage	0.285 (0.038)***	0.379 (0.052)***	0.358 (0.044)***	0.342 (0.039)***	0.325 (0.036)***	0.306 (0.036)***	0.282 (0.040)***	0.252 (0.051)***	0.217 (0.068)***	0.162 (0.097)*
Number of obs.	2197									
F-Value	85.84***									

Notes: In the table, the results of fixed effect (FE) covering the period 2000–2018 are provided with quantile regression results from 0.1 to 0.9 for distinct levels of investment (I/K) . Investments (I/K) are dependent variables, and explanatory variables are internal funds (CF/K) and external funds $(EXTF/K)$, respectively. Market-to-book (M/B) is the control variable. Additional control variables used in Panel B are firm size, namely $\ln(\text{sales})$, cash holdings (Cash/K) and Leverage. *, **, and *** show significance at 10%, 5%, and 1% levels, respectively.

The scalar coefficient $\alpha_i(\tau) \equiv \alpha_i + \delta_i q(\tau)$ is the quantile τ fixed effect for firm i or the distributional effect at τ . Differing from the usual fixed effect, the distributional effect represents the effect of time-invariant characteristics, which are allowed to have different impacts on the conditional distribution of investments of different firms. $\int_0^1 q(\tau) d\tau = 0$ implies that α_i can be interpreted as the average effect for firm i .

4. Empirical findings

In the first part of this section, descriptive statistics related to the data used in the study are presented. In the second and third parts, the test results of the first and second rung of the pecking order are respectively presented. In the fourth part, the test results of the measurements of the effect of the pecking order theory in large and small firms are shown. In the fifth part, the results related to the impact of the financial hierarchy from 2000 to 2009 and 2010–2018 are given. The results regarding the leverage/risk effect are presented in the sixth part. The last part includes the findings related to the subsectors.

4.1. Descriptive statistics

In Table 2, the descriptive statistics of the sample, which are ranked in equal groups according to the investment level (I/K) in the period from 2000 to 2018, are given in Fig. 1; and the graph related to the investment distribution of the firms is shown. Table 2 obviously shows that the total sample mean of the investments (I/K) is 0.278. When the investment variables as deciles are examined, the lowest (I/K) mean is 0.031 and the highest mean value in the tenth decile is 0.770 with a standard deviation of 0.136.

Considering the other explanatory variables used in the models, it was found that the mean values of the variable (CF/K) representing the first rung of the pecking order theory were higher than the mean values of the decile ($EXTF/K$) with lower standard deviations. This result indicates that the volatility in ($EXTF/K$) is more intense. When the variables are evaluated separately, the results indicate that there is an

increasing trend in the means and the standard deviations of (CF/K) and ($EXTF/K$). The means of the variable (CF/K) vary from 0.284 to 0.466, and the means of the variable ($EXTF/K$) vary from 0.111 to 0.351. When the variables ($DEBTF/K$) and ($EQUITYF/K$), which indicate the external financing alternatives used in the second rung of the pecking order, are analyzed according to deciles, the mean of ($DEBTF/K$) is higher than the mean of ($EQUITYF/K$). While the means of ($DEBTF/K$) have an increasing trend in the range of 0.0081–0.281 in terms of deciles, ($EQUITYF/K$) has an unstable mean in deciles with the highest mean of 0.069. When the control variables used in the models are analyzed, it is seen that the means of the $\ln(\text{sales})$ and leverage variables increased as the investment level increased until the seventh decile. This indicates that firms with increasing amounts of investment are turning to debts and that the share of debts within the capital structure has increased.

4.2. Panel quantile regression results for the first rung of pecking order

A summary of the results of the estimation for the first rung of the pecking order theory is available in Table 3 and Fig. 2. Here, the sensitivity of investments at distinct levels to internal funds (CF/K) and external funds ($EXTF/K$) is estimated by a quantile regression. Table 3 consists of two panels. Only the market-to-book ratio (M/B) was used as the control variable in Panel A; and firm size, namely, $\ln(\text{sales})$, cash holdings ($Cash/K$) and leverage, are included in the model as control variables in Panel B.

An examination of Panel A reveals that in all of the established models, the internal fund (CF/K) and external fund ($EXTF/K$) variables have statistically significant effects on investments. In addition, in all models, investments were found to be more sensitive to internal funds (CF/K) than external funds ($EXTF/K$). The fixed effect (FE) regression results estimated over the entire sample showed that internal funds had an impact of 0.090 and external funds had an effect of 0.054. When the quantile regression results are analyzed, it is seen that the effects of internal and external fund use

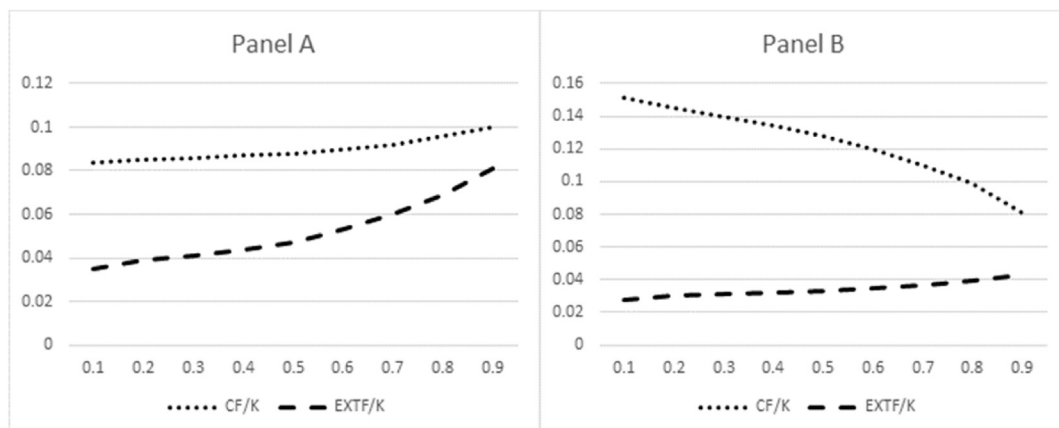


Fig. 2. Pecking order first rung quantile regression coefficients.

increase as the investment level increases. For example, in the 1st decile, the total effect on investments was 0.119 (0.084 + 0.035); whereas in the 9th decile, the total effect was 0.181 (0.100 + 0.081). In addition, when the relations of the internal and external funds with the investment level are examined, it is seen that both have an increasing slope (Fig. 2). However, the increasing slope of the use of external funds is higher than that for the use of internal funds. For example, according to results based on 1st and 9th deciles, the effect of using external funds to finance investment levels increased from 0.035 to 0.081 whereas the effect of using internal funds increased slightly from 0.084 to 0.100, respectively.

An examination of Panel B, where additional control variables were added, shows that the effect of using internal funds to finance investments was higher. However, when compared to the results obtained from Panel A, the effect has a decreasing slope as the investment level increases (Fig. 2). In addition, the effect of internal funds on investments is above 0.101 in all deciles except the 9th decile. Although the external funding coefficients are lower than those in the previous model, it is seen that the effect increases as the amount of investment increases. The coefficients are 0.028 at the 1st decile and 0.043 at the 9th decile. When these results are evaluated together, they confirm that the first rung of the pecking order is valid in manufacturing firms listed on the Borsa İstanbul. In other words, when businesses are making their investment financing decisions, the results show that the businesses primarily use internal funds followed by external funds. Moreover, although the increase in the investment level does not change this hierarchy, it shows that the sensitivity of investments to external funds increased relative to the sensitivity of investment to internal funds.

4.3. Panel quantile regression results for the second rung of pecking order

Summary data on the estimation results for the second rung of the pecking order theory are shown in Table 4 and Fig. 3. Here, the external financing variable ($EXTF/K$) used in the previous model is divided into two variables, namely, debt financing ($DEBTF/K$) and equity financing ($EQUITYF/K$). With this model, the general objective is to choose between the two external funding alternatives to determine the one that firms prefer in the second rung of their investment financing. Table 4 consists of two panels. In Panel A, only the market-to-book ratio (M/B) was used as the control variable; while in Panel B, firm size ($\log(\text{sales})$), cash holdings ($Cash/K$) and leverage were added to the model as control variables.

When the fixed effect (FE) regression results in Table 4 are examined, the results in Panel A and Panel B support the pecking order theory. In Panel A, the sensitivities of the internal funds (CF/K), debt financing ($DEBTF/K$) and equity financing ($EQUITYF/K$) to investments are 0.089, 0.062, and 0.027, respectively; while the coefficients in Panel B are 0.121, 0.037, and 0.029, respectively. When Panel A is analyzed, it is found that in all of the established models, the internal fund (CF/K) and debt financing ($DEBTF/K$) variables

Table 4
Pecking order second rung quantile regression coefficients.

Variables	FE	.10	.20	.30	.40	.50	.60	.70	.80	.90
Panel A: with M/B as a control variable										
CF/K	0.089 (0.019)***	0.084 (0.023)***	0.085 (0.020)***	0.085 (0.019)***	0.086 (0.017)***	0.087 (0.017)***	0.089 (0.019)***	0.091 (0.026)***	0.094 (0.038)***	0.098 (0.067)*
$DEBTF/K$	0.062 (0.011)***	0.038 (0.019)**	0.042 (0.017)**	0.045 (0.015)***	0.049 (0.014)***	0.054 (0.014)***	0.061 (0.016)***	0.071 (0.021)***	0.084 (0.031)***	0.099 (0.043)**
$EQUITYF/K$	0.027 (0.021)	0.036 (0.042)	0.034 (0.036)	0.033 (0.033)	0.031 (0.031)	0.030 (0.030)	0.027 (0.034)	0.024 (0.046)**	0.019 (0.067)	0.013 (0.094)
M/B	-0.002 (0.002)	0.001 (0.002)	0.001 (0.001)	0.000 (0.001)	-0.000 (0.001)	-0.001 (0.002)	-0.002 (0.001)	-0.004 (0.002)*	-0.006 (0.003)*	-0.009 (0.004)*
Number of Obs.	2197									
F-Value	15.860***									
Panel B: with additional control variable										
CF/K	0.121 (0.018)***	0.155 (0.027)***	0.147 (0.022)***	0.141 (0.020)***	0.135 (0.018)***	0.128 (0.018)***	0.120 (0.020)***	0.109 (0.026)***	0.096 (0.034)***	0.077 (0.049)
$DEBTF/K$	0.037 (0.011)***	0.012 (0.021)	0.018 (0.017)	0.022 (0.016)	0.026 (0.014)*	0.031 (0.014)**	0.037 (0.015)**	0.045 (0.021)**	0.055 (0.027)**	0.069 (0.039)*
$EQUITYF/K$	0.029 (0.019)	0.073 (0.034)	0.063 (0.028)**	0.056 (0.025)**	0.047 (0.023)**	0.039 (0.022)*	0.028 (0.027)	0.013 (0.033)	-0.002 (0.043)	-0.028 (0.063)
M/B	-0.004 (0.002)	0.000 (0.001)	-0.001 (0.001)	-0.002 (0.001)	-0.003 (0.001)**	-0.004 (0.001)**	-0.005 (0.001)***	-0.006 (0.002)***	-0.007 (0.003)***	-0.010 (0.004)**
$\log(\text{Sales})$	-0.108 (0.002)***	-0.086 (0.007)***	-0.091 (0.006)***	-0.095 (0.005)***	-0.099 (0.005)***	-0.010 (0.005)***	-0.108 (0.006)***	-0.115 (0.007)***	-0.123 (0.010)***	-0.136 (0.014)***
$Cash/K$	-0.046 (0.009)***	-0.036 (0.012)***	-0.039 (0.010)***	-0.040 (0.009)***	-0.042 (0.008)***	-0.044 (0.008)***	-0.047 (0.009)***	-0.050 (0.012)***	-0.053 (0.016)**	-0.059 (0.023)**
Leverage	0.282 (0.039)***	0.400 (0.053)***	0.372 (0.044)***	0.353 (0.039)***	0.331 (0.036)***	0.308 (0.036)***	0.279 (0.040)***	0.241 (0.051)***	0.198 (0.068)***	0.130 (0.097)
Number Of Obs.	2197									
F-Value	73.560***									

Notes: In the table, fixed effect (FE) regression results and quantile regression results of (I/K) were presented at different levels from 0.1 to 0.9. Investment (I/K) is the dependent variable and explanatory variables are internal funds (CF/K), debt financing ($DEBTF/K$) and equity financing ($EQUITYF/K$). Market-to-book (M/B) is the control variable. Additional control variables in Panel B are firm size ($\log(\text{sales})$), cash holdings ($Cash/K$) and Leverage. *, **, and *** show significance at 10%, 5%, and 1% levels, respectively.

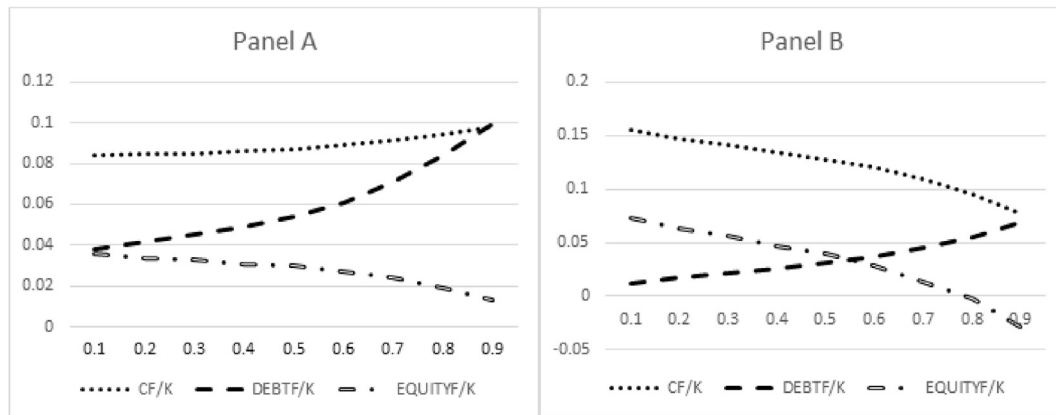


Fig. 3. Pecking order second phase quantile regression coefficients.

significantly affect investments. When the coefficients of the established models are examined, the results agree with the second rung of the pecking order. In other words, firms choose to use internal sources of funds (CF/K) first and then consider debt and equity financing in that order. According to the Panel A quantile regression results, the use of internal funds and debt financing to finance a firm's investment level increased together. The coefficient of internal fund (CF/K) usage increased from 0.084 at the 1st decile to 0.098 at the 9th decile; however, the debt financing ($DEBTF/K$) coefficient increased from 0.038 to 0.029 at these respective deciles. In addition, although the equity financing ($EQUITYF/K$) variable is statistically nonsignificant, it is obvious that the investment level increases. The coefficient of equity financing ($EQUITYF/K$) decreased from 0.036 at the 1st decile to 0.013 at the 9th decile.

The sensitivity of (CF/K) has a greater effect than varying the ($DEBTF/K$) and ($EQUITYF/K$) variables, and you may need to decrease it as the investment level increases. In addition, equity financing increased up to the 5th decile ($DEBTF/K$). However, after the 6th decile, the effect of debt financing ($DEBTF/K$) increases.

4.4. Pecking order effect in large and small firms

In the previous sections, it was found that firms follow pecking order in their investment financing. In this section, the present study investigates whether the effect of the pecking order changes financing preferences according to the size of the firms. The firms in the sample are classified based on their total assets. The firms in the largest 30% and the smallest 30% ranges are included in the analysis. Table 5 shows the fixed effect (FE) and quantile regression results. The results in Panel A show the first phase of the pecking order in large and small firms, and the results in Panel B show the first phase of the pecking order. When the results of Panel A are comparatively analyzed, they reveal that both large and small firms meet their investment needs with internal sources of funds. According to the fixed effect (FE) regression results, for large firms, the sensitivity of investments to internal funds (CF/K)

was 0.169, and the sensitivity to external funds was 0.028. For small firms, those numbers were 0.063 and 0.047, respectively. The quantile regression results show that both large and small firms prefer internal financing to external financing.

In Panel B, where the second rung of the pecking order was tested, the variables of debt financing ($DEBTF/K$) and equity financing ($EQUITYF/K$) were found to be statistically nonsignificant in the results of the large firm quantile regression. In addition, the effect of using internal funds (CF/K) to finance investments is significant in all models, and its coefficient is over 0.163. When the small firm results are analyzed, it is seen that debt financing ($DEBTF/K$) has a greater impact on investments in all models. This effect is 0.062, 0.077 and 0.107 at the 1st, 2nd and 3rd quantiles, respectively. When the results are generally evaluated, it is clearly seen that the sensitivity of large firm investments to internal funds (CF/K) is almost three times that of small firm investments (0.169 vs. 0.063, respectively). This indicates that large firms mostly use internal funds (CF/K). However, it is observed that borrowing is a priority in small firms, and its effect increases as the investment level increases.

4.5. Time horizon effect

This part of the study conducts an analysis to determine whether the effect of the pecking order changes by period. Periods are formed by dividing the sample period into two homogeneous equal groups, and the common features of the periods are the rising trend of the Borsa İstanbul Index. The differing aspects are that the local and global economic crisis occurred in the 2000–2009 period. The period from 2010 to 2018 covers the period after 2010 when the impact of the global economic crisis began to disappear.

Table 6 presents the fixed effect (FE) and quantile regression results for the pecking order effect in the periods from 2000 to 2009 and 2010–2018. When the Panel A and Panel B results for the period of 2000–2009 are analyzed together, the internal funds (CF/K), external funds ($EXTF/K$) and borrowing ($DEBTF/K$) variables were significant in all models. However, in all models, firms seem to prefer internal

Table 5
Large and small firm effect.

Variables	Large Firms				Small Firms			
	FE	.25	.50	.75	FE	.25	.50	.75
Panel A: Pecking Order First Rung								
<i>CF/K</i>	0.169 (0.043)***	0.176 (0.044)***	0.172 (0.038)***	0.163 (0.069)**	0.063 (0.034)*	0.062 (0.034)*	0.063 (0.030)**	0.064 (0.052)
<i>EXTF/K</i>	0.028 (0.017)*	0.010 (0.024)	0.021 (0.021)	0.042 (0.038)	0.047 (0.002)***	0.043 (0.027)	0.045 (0.061)*	0.050 (0.042)
<i>M/B</i>	0.022 (0.006)	0.021 (0.006)***	0.022 (0.006)***	0.023 (0.010)**	−0.005** (0.002)	−0.004 (0.001)**	−0.005 (0.001)***	−0.006 (0.002)***
<i>Number Of Obs.</i>	636				522			
<i>F-Value</i>	12.180***				6.220***			
B: Pecking Order Second Rung								
<i>CF/K</i>	0.169 (0.043)***	0.176 (0.043)***	0.172 (0.038)***	0.163 (0.069)**	0.063 (0.034)**	0.064 (0.034)*	0.063 (0.029)**	0.062 (0.052)
<i>DEBTF/K</i>	0.030 (0.017)*	0.014 (0.024)	0.023 (0.021)	0.042 (0.038)	0.088 (0.027)***	0.062 (0.037)*	0.077 (0.032)**	0.107 (0.056)*
<i>EQUITYF/K</i>	−0.012 (0.102)	−0.096 (0.150)	−0.047 (0.002)	0.050 (0.238)	0.007 (0.026)	0.030 (0.001)	0.017 (0.033)	−0.008 (0.058)
<i>M/B</i>	0.022 (0.006)***	0.021 (0.006)***	0.022 (0.006)***	0.023 (0.010)**	−0.005 (0.003)***	−0.004 (0.001)**	−0.005 (0.001)***	−0.006 (0.010)**
<i>Number of obs.</i>	636				522			
<i>F-Value</i>	15.540***				5.510***			

Notes: In the table, quantile regression results for different levels of investment (*I/K*) (0.25, 0.50 and 0.75) are given together with the fixed effect (*FE*) regression results covering the period 2000–2018. Investments (*I/K*) are dependent variables, and respectively explanatory variables are internal funds (*CF/K*), debt financing (*DEBTF/K*) and equity financing (*EQUITYF/K*). Market-to-book (*M/B*) is the control variable. *, ** and *** show significance at the 10%, 5% and 1% levels.

Table 6
Time horizon effect.

Variables	2000–2009 Period				2010–2018 Period			
	FE	.25	.50	.75	FE	.25	.50	.75
Panel A: Pecking Order First Rung								
<i>CF/K</i>	0.181 (0.035)***	0.222 (0.038)***	0.196 (0.031)***	0.143 (0.049)***	−0.004 (0.022)	0.008 (0.027)	−0.000 (0.023)	−0.015 (0.038)
<i>EXTF/K</i>	0.058 (0.014)***	0.054 (0.024)**	0.056 (0.021)***	0.061 (0.032)*	0.050 (0.012)***	0.033 (0.017)*	0.045 (0.014)***	0.067 (0.024)***
<i>M/B</i>	0.002 (0.006)	0.006 (0.005)	0.003 (0.004)***	−0.000 (0.007)	−0.002 (0.002)	−0.001 (0.001)	−0.001 (0.001)	−0.002 (0.002)
<i>Number of obs.</i>	1027				1170			
<i>F-Value</i>	17.010***				5.480***			
Panel B: Pecking Order Second Rung								
<i>CF/K</i>	0.181 (0.035)***	0.221 (0.038)***	0.195 (0.031)***	0.142 (0.051)***	−0.006 (0.034)	0.004 (0.027)	−0.003 (0.023)	−0.016 (0.038)
<i>DEBTF/K</i>	0.075 (0.019)***	0.065 (0.027)**	0.071 (0.001)***	0.084 (0.036)**	0.060 (0.013)***	0.043 (0.017)**	0.055 (0.015)***	0.076 (0.056)***
<i>EQUITYF/K</i>	0.020 (0.030)	0.045 (0.046)	0.029 (0.038)	−0.004 (0.062)	−0.024 (0.036)	−0.000 (0.043)	−0.016 (0.037)	−0.045 (0.061)
<i>M/B</i>	0.002 (0.005)	0.006 (0.005)	0.003 (0.004)	−0.001 (0.007)	−0.002 (0.002)	−0.001 (0.001)	−0.001 (0.001)	−0.002 (0.002)
<i>Number Of Obs.</i>	1027				1170			
<i>F-Value</i>	13.260***				5.370***			

Notes: In the table, the results of fixed effect (*FE*) covering the period of 2000–2018 are provided with quantile regression results for different levels of investment (*I/K*) (0.25, 0.50 and 0.75). Investments (*I/K*) are dependent variables, and explanatory variables are internal funds (*CF/K*), debt financing (*DEBTF/K*) and equity financing (*EQUITYF/K*). Market-to-book (*M/B*) is the control variable. *, ** and *** show significance respectively at the 10%, 5% and 1% levels.

funds (CF/K) over ($EXTF/K$), ($DEBTF/K$) and ($EQUITYF/K$) in investment financing. The sensitivity of investments to internal funds (CF/K) is 0.222, 0.196 and 0.143 at the 1st, 2nd and 3rd quantiles, respectively. The results also provide evidence that firms prefer debt financing ($DEBTF/K$) to equity financing in the second rung of the pecking order. The effect of debt financing ($DEBTF/K$) on investments increased from 0.065 to 0.084 as the investment level increased. The results of the analysis for the period from 2000 to 2009 support the financial hierarchy effect. When the results covering the period from 2010 to 2018 are analyzed, external funds ($EXTF/K$) variable in Panel A and the debt financing ($DEBTF/K$) variable in Panel B were found to be statistically significant in all models. In both models, the effects of these variables on investments increased. The external fund ($EXTF/K$) variable had the highest value at the 3rd quantile with 0.067, and the debt financing ($DEBTF/K$) variable had the highest value of 0.076.

When the results of the analysis for both periods are examined, it indicates that the periods have a significant effect on financing preferences. In other words, while it was seen that firms were following the pecking order process in the search for funds in the period from 2000 to 2009, it was revealed that they preferred borrowing over other alternatives in the period from 2010 to 2018.

4.6. Leverage/risk effect

In this section, the effect of the difference in the target leverage of firms on pecking order behaviour is investigated. During the analysis, firms with leverage rates above 60% are classified as high risk/leveraged firms, and those with rates below 40% are classified as low risk/leveraged firms. The results of the analysis presented in Table 7 reveal that in the first rung of the pecking order, high-risk/leveraged firms preferred internal financing (CF/K) to external financing ($EXTF/K$); and as the investment level increased, the sensitivity of internal financing at the 1st, 2nd and 3rd quantiles increased to 0.166, 0.181 and 0.212, respectively, in all of the models. However, the effect of external financing ($EXTF/K$) in firms with low leverage is statistically significant at the 1st and 2nd quantiles with the fixed effect model, and its sensitivity to investments is higher compared to internal financing (CF/K).

When the second rung of the pecking order is examined, it is revealed that low target leverage firms prefer debt financing ($DEBTF/K$) to stock issuance ($EQUITYF/K$). While debt financing was statistically significant at the 1st and 2nd quantile investment levels at 0.099 and 0.076, respectively, at 1% significance level, no significant relationship was found in the share issuances ($EQUITYF/K$). It was found that highly leveraged firms preferred internal financing (CF/K) as the first alternative, and all models were statistically significant with high coefficients (1st quantile, 0.167; 2nd quantile, 0.187; and 3rd quantile, 0.224). In addition, it was observed that debt financing ($DEBTF/K$) did not have a statistically significant effect on changes in investment levels. A statistically significant relationship was found in the fixed effect model for the

Variables	Low Leverage Firms			High Leverage Firms		
	.25	.50	.75	.25	.50	.75
Panel A: Pecking Order First Rung						
CF/K	0.043 (0.027)*	0.047 (0.024)**	0.045 (0.022)**	0.040 (0.047)	0.189 (0.051)***	0.212 (0.075)***
$EXTF/K$	0.047 (0.020)**	0.060 (0.023)***	0.054 (0.021)***	0.036 (0.043)	0.031 (0.016)**	0.051 (0.033)
M/B	-0.026 (0.006)	-0.013 (0.005)**	-0.020 (0.005)***	0.038 (0.011)***	0.001** (0.003)	0.001 (0.004)
Number Of Obs.	1002				521	
F-Value	9.990***				8.330***	
Panel B: Pecking Order Second Rung						
CF/K	0.043 (0.027)*	0.044 (0.030)	0.043 (0.027)*	0.041 (0.051)	0.197 (0.051)***	0.224 (0.074)***
$DEBTF/K$	0.055 (0.041)	0.099 (0.032)***	0.076 (0.028)***	0.017 (0.054)	0.028 (0.016)*	0.046 (0.034)
$EQUITYF/K$	0.038 (0.044)	0.018 (0.046)	0.028 (0.041)	0.056 (0.078)	0.155 (0.082)*	0.242 (0.135)*
M/B	-0.026 (0.007)***	-0.013 (0.007)*	-0.020 (0.006)***	0.038 (0.012)***	0.001 (0.003)	0.001 (0.004)
Number Of Obs.	1002				521	
F-Value	4.510***				6.540***	

Notes: In the table, the results of fixed effect (FE) covering the period of 2000–2018 are provided with quantile regression results for different levels of investment (I/K) (0.25, 0.50 and 0.75). Investments (I/K) are dependent variables, and explanatory variables are internal funds (CF/K), debt financing ($DEBTF/K$) and equity financing ($EQUITYF/K$). Market-to-book (M/B) is the control variable. *, ** and *** show significance respectively at the 10%, 5% and 1% levels.

Table 8
Sub-sector effect.

Variables	Panel A: Pecking Order First Rung				Panel B: Pecking Order Second Rung			
	FE	.25	.50	.75	FE	.25	.50	.75
Sub-sectors: Food, Drink and Tobacco Products Manufacturing								
CF/K	0.160 (0.018)**	0.164 (0.083)**	0.1610 (0.070)**	0.157 (0.117)	0.157 (0.069)**	0.173 (0.078)**	0.162 (0.066)**	0.145 (0.115)
EXTF/K	0.128 (0.028)***	0.096 (0.037)***	0.120 (0.031)***	0.156 (0.052)***				
DEBTF/K					0.167 (0.031)***	0.143 (0.039)***	0.159 (0.033)***	0.188 (0.058)***
EQUITYF/K					−0.035 (0.060)	−0.024 (0.053)	−0.031 (0.044)	−0.044 (0.077)*
M/B	−0.002 (0.006)	0.001 (0.006)	−0.001 (0.005)	−0.005 (0.009)	−0.001 (0.006)	0.003 (0.006)	0.001 (0.005)	−0.005 (0.008)
Number Of Obs.	296				296			
F-Value	8.090***				8.560***			
Sub-sectors: Weaving, Clothing and Leather Products Manufacturing								
CF/K	0.115 (0.061)*	0.075 (0.071)	0.102 (0.061)*	0.147 (0.100)	0.093 (0.061)	0.065 (0.068)	0.083 (0.058)	0.116 (0.098)
EXTF/K	0.038 (0.025)	0.032 (0.026)	0.036 (0.023)	0.043 (0.038)***				
DEBTF/K					0.030 (0.026)	0.026 (0.026)	0.029 (0.022)	0.033 (0.037)
EQUITYF/K					0.264 (0.105)**	0.294 (0.078)***	0.275 (0.066)***	0.240 (0.113)**
M/B	0.017 (0.017)	0.015 (0.020)	0.017 (0.016)	0.019 (0.027)	0.019 (0.018)	0.015 (0.020)	0.018 (0.017)	0.023 (0.029)
Number Of Obs.	273				273			
F-Value	3.100**				3.570***			
Sub-sectors: Forest and Forest Products Manufacturing								
CF/K	0.061 (0.052)	0.057 (0.048)	0.059 (0.042)	0.065 (0.078)	0.061 (0.052)	0.057 (0.049)	0.059 (0.043)	0.062 (0.078)
EXTF/K	0.300 (0.065)***	0.183 (0.091)**	0.240 (0.082)***	0.402 (0.154)***				
DEBTF/K					0.309 (0.066)***	0.193 (0.095)**	0.251 (0.084)***	0.407 (0.155)***
EQUITYF/K					0.087 (0.312)	−0.132 (0.417)**	−0.023 (0.365)	0.270 (0.667)
M/B	0.027 (0.015)*	0.032 (0.016)*	0.029 (0.014)**	0.023 (0.027)	0.027 (0.015)*	0.031 (0.016)	0.029 (0.014)**	0.023 (0.026)
Number Of Obs.	217				217			
F-Value	9.680***				7.370***			
Sub-sectors: Chemical, Petroleum, Rubber and Plastic Products Manufacturing								
CF/K	0.099 (0.050)**	0.100 (0.050)**	0.100 (0.047)**	0.099 (0.076)	0.106 (0.050)**	0.105 (0.051)**	0.106 (0.046)**	0.107 (0.078)
EXTF/K	0.085 (0.034)***	0.085 (0.028)***	0.085 (0.024)***	0.085 (0.042)**				
DEBTF/K					0.076 (0.033)**	0.084 (0.027)***	0.079 (0.024)***	0.070 (0.041)*
EQUITYF/K					0.254 (0.168)	0.246 (0.157)	0.026 (0.138)*	0.260 (0.236)
M/B	0.017 (0.009)**	0.018 (0.008)**	0.018 (0.007)**	0.019 (0.013)	0.019 (0.009)**	0.018 (0.008)**	0.018 (0.007)**	0.020 (0.013)
Number Of Obs.	356				356			
F-Value	6.390***				5.050***			
Sub-sectors: Stone and Soil Based Industrial Products Manufacturing								
CF/K	0.207 (0.047)***	0.150 (0.053)***	0.174 (0.048)***	0.242 (0.093)***	0.209 (0.047)***	0.158 (0.051)***	0.181 (0.046)***	0.244 (0.094)***
EXTF/K	0.162 (0.067)***	0.137 (0.078)*	0.148 (0.070)**	0.178 (0.135)				
DEBTF/K					0.147 (0.068)**	0.133 (0.076)*	0.139 (0.068)**	0.156 (0.139)
EQUITYF/K					0.460 (0.251)*	0.313 (0.273)	0.378 (0.246)	0.561 (0.498)
M/B	−0.021 (0.025)	−0.001 (0.007)	−0.009 (0.007)	−0.035 (0.014)**	−0.025 (0.009)***	−0.001 (0.008)	−0.012 (0.007)	−0.041 (0.016)**
Number Of Obs.	445				445			
F-Value	8.820***				7.000***			
Sub-sectors: Metal Main Industry Products Manufacturing								
CF/K	0.014 (0.063)	0.045 (0.074)	0.022 (0.062)	−0.013 (0.104)	0.012 (0.063)	0.042 (0.077)	0.021 (0.207)	−0.014 (0.616)
EXTF/K	0.113 (0.037)***	0.065 (0.065)	0.100 (0.055)*	0.155 (0.091)*				
DEBTF/K					0.151 (0.046)***	0.128 (0.071)*	0.144 (0.190)	0.173 (0.568)
EQUITYF/K					0.060 (0.054)	0.044 (0.084)	0.055 (0.226)	0.075 (0.673)
M/B	−0.019 (0.011)*	−0.009 (0.008)	−0.016 (0.006)**	−0.028 (0.011)**	−0.020 (0.011)*	−0.010 (0.008)	−0.017 (0.021)	−0.030 (0.064)
Number Of Obs.	236				236			
F-Value	3.450**				3.030**			
Sub-sectors: Metal Goods and Machinery Products Manufacturing								
CF/K	0.037 (0.035)	0.053 (0.034)	0.044 (0.030)	0.024 (0.054)	0.039 (0.035)	0.057 (0.035)*	0.047 (0.030)	0.024 (0.055)
EXTF/K	0.023 (0.012)*	0.029 (0.014)*	0.026 (0.012)**	0.019 (0.023)*				
DEBTF/K					0.017 (0.018)***	0.016 (0.017)	0.017 (0.015)	0.018 (0.028)
EQUITYF/K					0.034 (0.026)	0.058 (0.029)**	0.045 (0.026)*	0.014 (0.047)
M/B	−0.004 (0.002)*	−0.002 (0.001)*	−0.004 (0.001)***	−0.007 (0.002)***	−0.004 (0.002)*	−0.002 (0.001)*	−0.004 (0.001)***	−0.006 (0.002)***
Number Of Obs.	374				374			
F-Value	2.810**				2.150*			

Notes: In the table, the results of fixed effect (FE) covering the period of 2000–2018 are provided with quantile regression results for different levels of investment (I/K) (0.25, 0.50 and 0.75). Investments (I/K) are dependent variables, and explanatory variables are internal funds (CF/K), debt financing (DEBTF/K) and equity financing (EQUITYF/K). Market-to-book (M/B) is the control variable. *, ** and *** show significance at the 10%, 5% and 1% levels respectively.

stock issuance ($EQUITYF/K$) variable at the 3rd quantile at the 10% significance level with coefficients of 0.155 and 0.242, respectively. When the results are generally evaluated, it can be seen that low- and high-leverage firms do not fully exhibit pecking order behaviour. Low-leverage firms are in line with the second rung of the pecking order while high-leverage firms are in line with the first rung of the pecking order.

4.7. Pecking order effect in manufacturing industry sub-sectors

In this part of the study, the pecking order effect in seven subsectors of the manufacturing industry was investigated. Table 8 presents the fixed effect (FE) and quantile regression results. The results in Panel A show the first rung of the pecking order in the manufacturing industry subsectors, and the results in Panel B show the second rung of the pecking order. When the Panel A results are comparatively analyzed, food, drink and tobacco product manufacturing; weaving, clothing and leather product manufacturing; chemical, petroleum, rubber and plastic product manufacturing; and stone- and soil-based industrial product manufacturing are among the subsectors with a higher sensitivity to internal funds (CF/K) than external funds ($EXTF/K$) when financing their investments. In addition, the sensitivity of investments to internal funds in the stone- and soil-based industrial product manufacturing subsector is 0.150, 0.174 and 0.242 at the 1st, 2nd and 3rd quantiles, respectively; and it was found that both the use of internal funds and the use of external funds increased as the investment level increased. In addition, it is observed that firms prefer external funds ($EXTF/K$) as the first option in investment financing in the forestry and forestry product manufacturing, metal main industrial product manufacturing and metalware and machinery manufacturing subsectors. Additionally, it was observed that the sensitivity to investments in the metalware and machinery product manufacturing subsector was lower compared to those of other subsectors.

When the analysis results of the second rung of the pecking order in the manufacturing industry subsectors as presented in Panel B of Table 8 are examined, it can be seen that firms in the food, drink and tobacco products manufacturing; chemical, petroleum, rubber and plastic products manufacturing; and stone- and soil-based industrial products manufacturing subsectors exhibit pecking order behaviour. While the sensitivity of internal funds (CF/K) to increases in the investment level in the food, drink and tobacco products manufacturing subsector decreased, the sensitivity of debt ($DEBTF/K$) in the subsector increased. In addition, although the coefficient of equity financing ($EQUITYF/K$) in the stone- and soil-based industrial product manufacturing subsector is above 0.30 in all models, it is statistically insignificant. When the results for the forestry and forestry product manufacturing and metal base industrial product manufacturing subsectors were examined, it was found that the debt variable ($DEBTF/K$) was the most sensitive variable to the change in investment levels. When the two

subsectors are compared in terms of quantiles, it is seen that as the investment level in the forest and forest product manufacturing subsector increases, the sensitivity to borrowing increases up to 0.40; furthermore, that of the metal main industrial product manufacturing subsector also increases, but it is statistically nonsignificant at the 2nd (0.144) and 3rd (0.173) quantiles. Except for the equity financing ($EQUITYF/K$) variable in the weaving, clothing and leather products manufacturing subsector, the other variables in the models were found to be statistically nonsignificant. The results show that an increase in the investment level in this sector leads firms to equity financing. In addition, a similar result is found for metal goods and machinery product manufacturing. However, the coefficients are relatively lower.

5. Concluding remarks

This study investigated the financial preferences of firms at different investment levels and tested the pecking order theory for firms within the Turkish manufacturing industry listed on the Borsa İstanbul. In this study, our focus was on the investment levels at which firms resort to internal funds, after which investment level they turn to external funds, and from which source (debt or equity) the external fund needs are first met. In this study, the quantile regression method was used to determine the relative importance of internal and external fund sources in the financing of firm investments. The study provides evidence that the pecking order theory is valid in manufacturing firms listed on the Borsa İstanbul. The results show that in the first rung of the pecking order, businesses invest primarily in internal funds and then in external funds. Firms' usage of internal funds mainly supports the argument given by Mayer (1990) that internal funds are a dominant source of funding for all firms and that the majority of firms are sensitive to existing liquidity in their investment decisions. In addition, as the investment level increases, the ranking order does not change, and the increase in the sensitivity of investments to internal and external funds also supports the view of Jensen (1986) that firms turn to new investments in order to evaluate existing internal funds. In the second rung of the pecking order, the empirical results demonstrated that the firms chose to borrow more after using their internal funds, and the borrowing effect increased as the investment level increased.

Another result of the study is that large firms are more sensitive to internal funds in the first rung of the pecking order than small firms. In addition, contrary to the common opinion in the literature (Cotei & Farhat, 2009; De Jong, Verbeek, & Verwijmeren, 2010; Lemmon & Zender, 2010), evidence has been found that small firms follow the pecking order. Additionally, it was found that the pecking order behaviours differed in periods. The empirical findings revealed that firms acted in accordance with both the first and the second rungs of pecking order in the 2000–2009 period; however, when financing their investments in the 2010–2018 period, they turned towards external funding/debt financing alternatives. One can argue that instability, especially the instability caused

by the economic and political crises experienced during the period from 2000 to 2009 (the 2001 crisis and the 2008 global economic crisis), caused firms to exhibit pecking order behaviour in line with the empirical findings of Köksal and Orman (2015). In this study, it is observed that firms with high leverage primarily use internal funds, and the sensitivity of internal funds increases as the investments increase. Besides, it was observed that these firms turned to share issuances due to the high level of leverage and the increase in the investment level. In low-leverage firms, borrowing was found to be more sensitive to investments. In either case, the results do not conform to the pecking order theory. Finally, the estimation results demonstrate that firms in the food, drink and tobacco product manufacturing; chemical, petroleum, rubber and plastic product manufacturing; and stone- and soil-based industrial product manufacturing subsectors showed pecking order behaviour.

Developing economies with their different set of characteristics has been growing rapidly in the last 20 years, and this study contributes to the existing literature due to its application of the theory in a developing economy context. Additionally, using fixed effects in the location-scale approach as introduced by Machado and Silva (2019) and considering company-specific effects in solving the endogeneity problem in the investment quantile regression model developed by Chay et al. (2015) were important.

In addition, the issuance of bonds as an external source of funds is not preferred by firms listed on the Borsa İstanbul. However, it is a structural problem of the country's market, and it is seen as a limitation of the study since it does not address that source of funding and the target debt capacity in the financial hierarchy. The present study was performed within a limited time period, and emphasis is placed on firms listed on the Borsa İstanbul. Additionally, Chay et al.'s (2015) quantile regression framework does not take into account the endogeneity of financing activities or the simultaneity among investment and financing activities due to some difficulties in identifying instruments for the financing variables. Nevertheless, Machado and Santos Silva (2019) proposed a useful quantile regression approach that estimates quantiles from estimates of the conditional mean and of the conditional scale function, which provides an easy method to estimate regression quantiles for circumstances when the use of the traditional quantile regression is difficult. This study also follows Machado and Santos Silva's (2019) approach to estimate quantiles. The fixed effects account for some endogeneity in the sense that they account for omitted variables that are constant over time and that can be correlated with the regressors. As far as is known, it is not possible to account for simultaneity in the quantile regression with fixed effects. Chay et al. (2015) also argue that the dynamic relation would require the estimation of a system of equations; these extensions of the current methodology can be assessed in future research since the dynamic quantile regression technique is still in the early stage of its development. The studies will help to explain why the pecking order effect differs from time to time with large and small firms growing fast in the future and

how it will contribute to the existing literature. In addition, the results of this study will provide firm managers, investors, credit institutions and academics with knowledge about the financial behaviour of manufacturing firms listed on the Borsa İstanbul. Since the pecking order theory is based on asymmetric information problems between managers and shareholders, the financing decisions made by managers send signals to shareholders and lenders. This study will help investors and credit institutions to predict which source of funding the firms in the relevant market and subsectors will apply should the need for financing arise and can especially contribute to the early positions of investors. In addition to offering better positioning opportunities to company managers according to the classification made in the study, it may also help them to analyze the funding of both their firms and their competitors. Furthermore, academicians can contribute to the development of this study by applying a different model to a developing economy and by creating new ideas about the pecking order theory that is currently being discussed.

Declaration of Competing Interest

There is no conflict of interest.

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